Your most important price is one you don't set.

The active pharmaceutical ingredient — the cost inside every medicine you make. **44 years of it in the US, 16 in Türkiye**, from the statistical agencies. Deliberately **costs only**: selling prices are shaped by regulators and rebates, so we measured the side nobody administers.

BASIS · FACTORY-GATE PRODUCER PRICE INDEXES, BLS & TÜİK · INPUT SIDE ONLY — WHY, ON PAGE 3 · HUMAN + VETERINARY PHARMA TOGETHER

Three facts about your cost of goods. All defensible.

UNITED STATES · 1982–2026

−44.8%

The ingredient's real price over 44 years — and **still falling**: −17.9% since 2020. A four-decade cost tailwind.

UNITED STATES · VOLATILITY

1.37×

The ingredient index moves **1.37× more** month-to-month than the finished-medicine index. The swings live on your cost side.

TÜRKIYE · 2010–2026

68%

Share of a lira move in your imported input cost **the same month** — and ~100% by the next. No warning, no lag.

ANCHOR

Price reviews and procurement to the **input curve** — it moves first and hardest.

HEDGE

Before the move, rolling quarter — pass-through completes in a month; a hedge arranged after is a receipt.

WATCH

The falsifier: **8 straight quarters of real ingredient increases** would end the tailwind era.

Selling prices lie. Input prices don't.

We deliberately excluded the output side — in every market, a published "selling price" is shaped by something other than the market:

TÜRKIYE

Human drug prices are **administered** — reference pricing under a state decree, converted at a **government-set euro rate** adjusted in periodic jumps, plus SGK reimbursement.

UNITED STATES

Published indexes are **list-basis** — PBM rebates on branded drugs mean net realisation can sit far below list, and the gap itself changes over time.

EUROPEAN UNION

Reference pricing, reimbursement negotiation and clawbacks — the published price is the regulated one, not the market one.

The input side has none of this machinery. Bulk ingredient and import prices are business-to-business market prices — no decree, no rebate, no reimbursement. That is the side this study measures, and the reason it can be defended line by line.

What exactly is being measured.

Producer price index

The average price **at the factory gate** — what manufacturers actually receive or pay, as collected by the statistical agency.

US ingredient =

Bulk active-ingredient manufacturing, NAICS 325411 medicinal & botanical — BLS PCU325411325411, monthly, Jun 1982 → Jun 2026, 529 observations.

TR imported input =

TÜİK foreign producer price index, pharmaceutical products — TP.UFEYD16, monthly, 2010 → 2026. An import price: world markets × the lira.

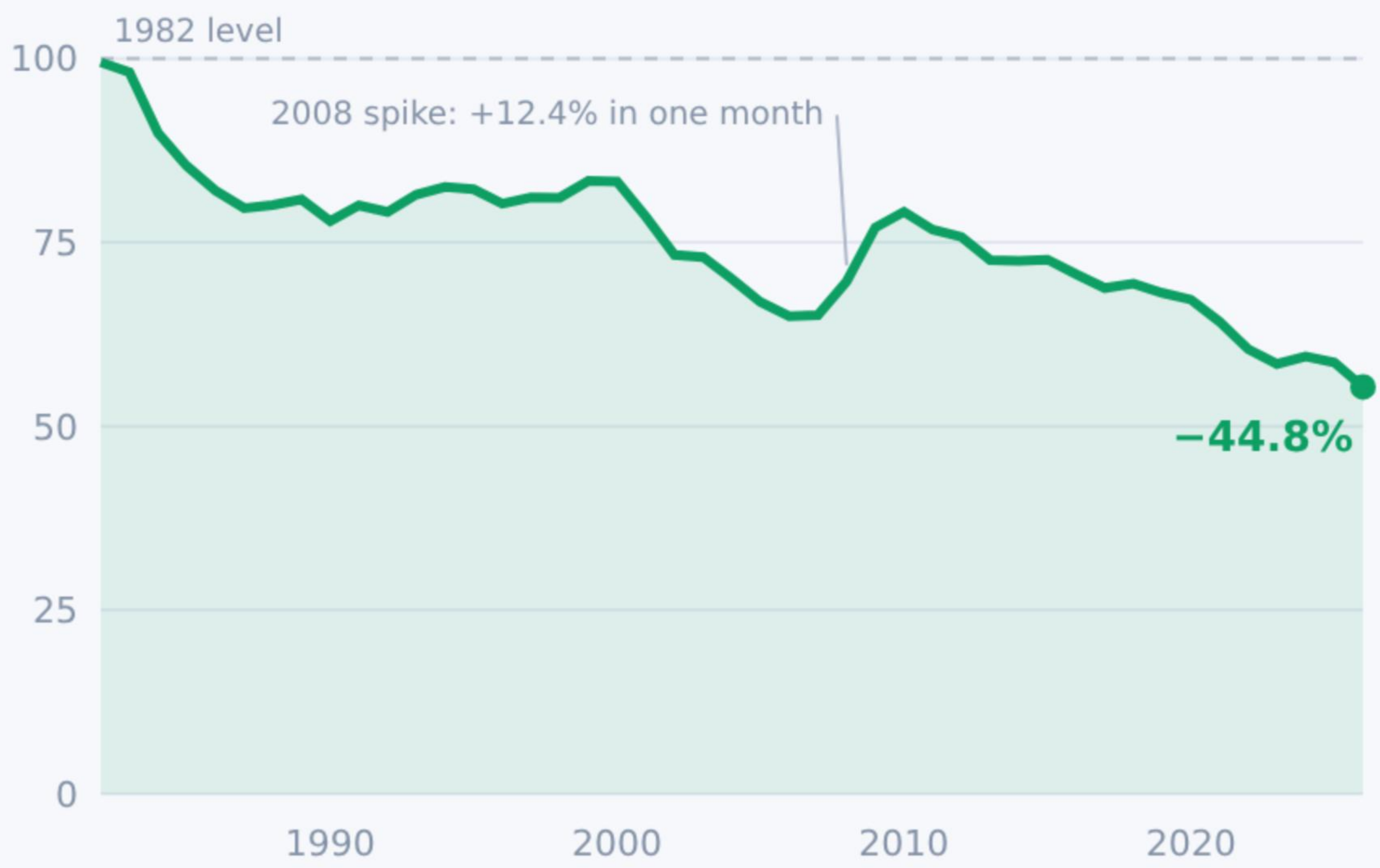
"Real" =

US series deflated by **CPI-U**. Turkish series kept nominal and used only in **ratios and monthly changes**, where inflation cancels. Only changes between named dates are quoted — never index levels.

HUMAN + VETERINARY TOGETHER — NO VETERINARY SPLIT EXISTS AT THE INGREDIENT STAGE (THE US FINISHED SIDE HAS ONE: WPU0634, TREATED SEPARATELY)

The ingredient fell 44.8% in real terms — and hasn't stopped.

Forty-four years of the US bulk active-ingredient price, inflation-adjusted. Down **−17.9% since 2020** alone. **Every margin plan in this industry rode this line down** — mostly without naming it.



US ONLY · JUN 1982 – JUN 2026 · BLS PCU325411325411 · CPI-U DEFLATED · JUN 1982 = 100 · ANNUAL MEANS OF MONTHLY DATA

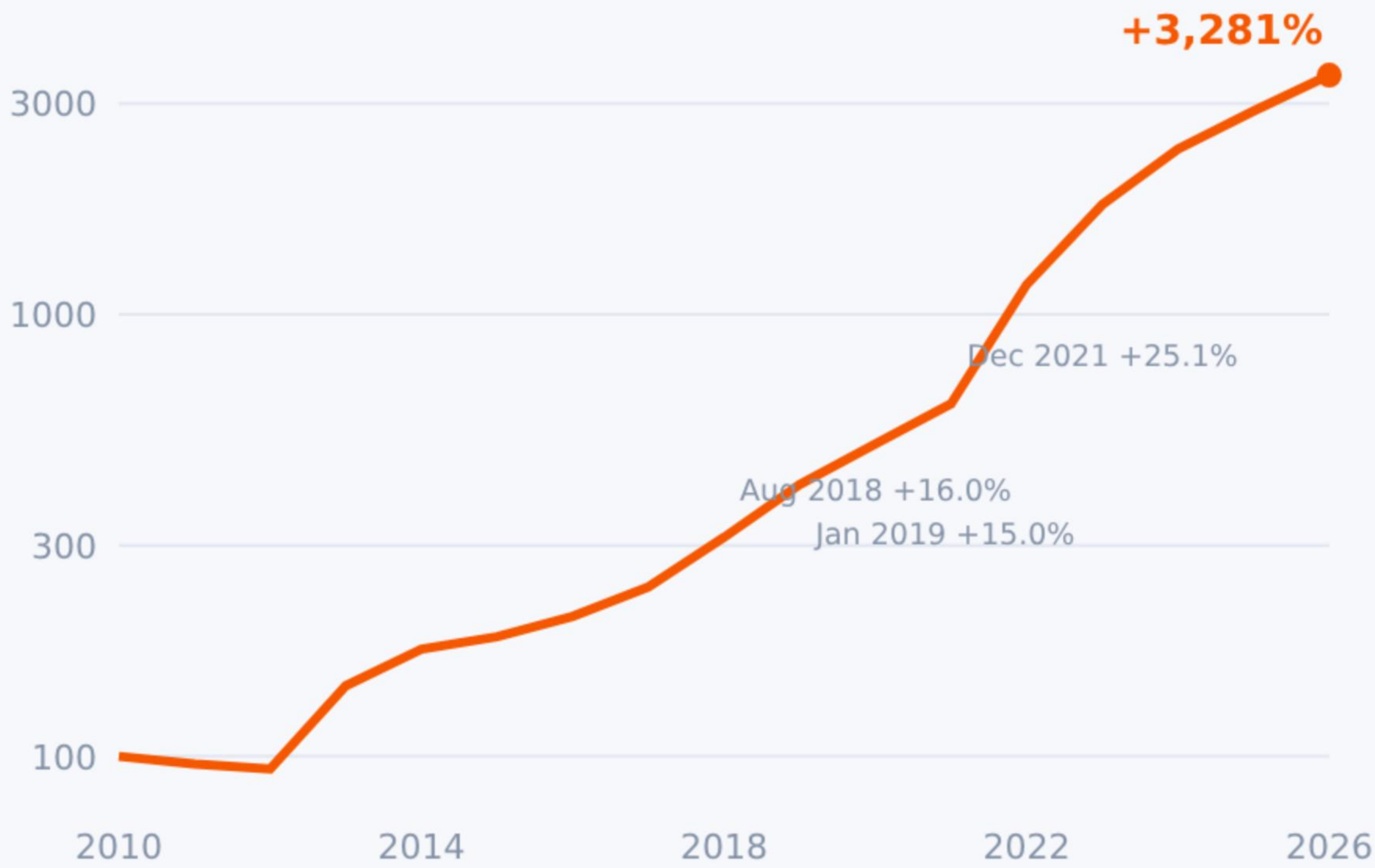
The swings live on the cost side.

Month-to-month, the ingredient index moves **1.37× more** than the finished-medicine index. Largest single month: **+12.4% in June 2008**. Most recent large move: **−4.4% in January 2026**. **A team watching its own selling price is watching the calm side of its own ledger.**

US ONLY · STDEV OF MONTH-ON-MONTH LOG CHANGES, JUN 1982 – JUN 2026, N = 527 · INGREDIENT 0.0102 VS FINISHED 0.0074 · NOMINAL INDEXES

The imported input rose 34-fold.

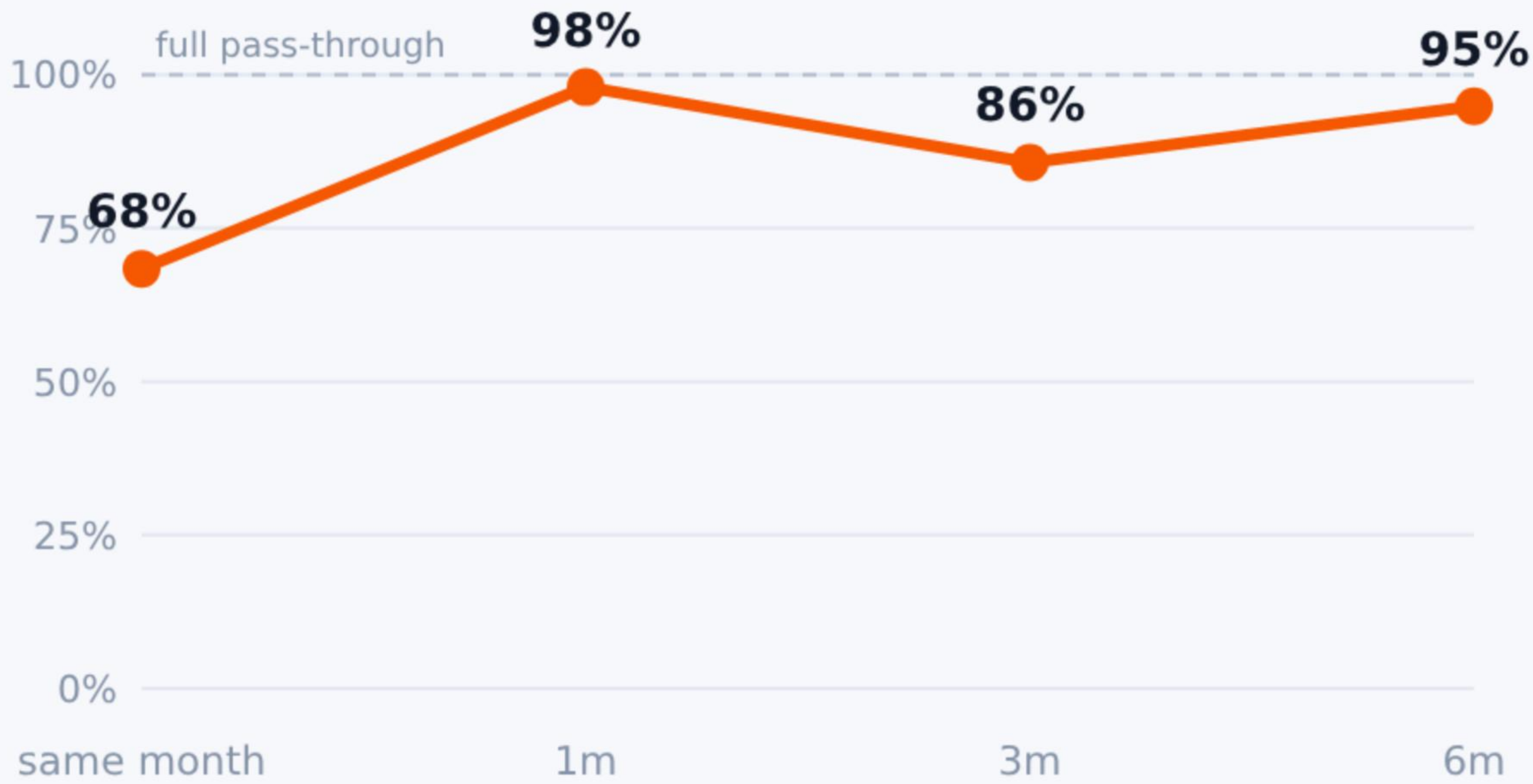
Türkiye's imported pharmaceutical input price, 2010 → 2026, log scale — **+3,281% in lira terms**. Its three largest monthly jumps are **lira dates, not pharma dates**: Dec 2021 +25.1%, Aug 2018 +16.0%, Jan 2019 +15.0%.



TÜRKİYE ONLY · 2010–2026 · TÜİK TP.UFEYD16 · NOMINAL TRY, 2010 = 100, LOG SCALE · LEVEL NEVER READ AS REAL GROWTH

A lira move is in your costs within one month.

Per 1% of USD/TRY: **0.68% lands the same calendar month, ~1.0% by the next.** The lag profile peaks at zero and collapses after — **there is no warning period to watch for.** Hedges work only if they exist before the move.



TÜRKİYE ONLY · 2010–2026 · Β = OLS ON MONTH-ON-MONTH LOG CHANGES, N = 197 · FX: OECD CCUSMA02TRM618N MONTHLY AVG · CUMULATIVE BY LOCAL PROJECTION

Manage the ledger you can defend: the input side.

ANCHOR

Tie price reviews and procurement to the input curve — it is 1.37× more volatile than the output side and moves first.

HEDGE

Rolling quarter, arranged in advance — pass-through completes within a month of the currency move.

WATCH

Eight straight quarters of real ingredient increases ends the 44-year tailwind. Nothing else on this page changes before that does.

Every source

US · BLS producer price index via FRED, monthly — PCU325411325411, medicinal & botanical manufacturing (bulk active ingredient), index Jun 1982 = 100, Jun 1982 → Jun 2026 · deflator CPIAUCSL (CPI-U)

TR · TÜİK via CBRT EVDS, monthly — TP.UFEYD16, foreign producer price index, pharmaceutical products, 2010 → 2026 · FX: OECD CCUSMA02TRM618N, USD/TRY monthly average of daily rates

Excluded by design · all selling-price series — administered in TR (pricing decree, SGK), list-basis in the US (rebates), reference-priced in the EU. Those findings live on the full study pages, with the caveats printed.

Every figure recomputed from data committed in the repository. Human + veterinary measured together; no veterinary split exists at the ingredient stage (the US finished side has WPU0634, treated separately).

NEXT — PART 2 · The powder is not the price. What does a medicine's price actually pay for — R&D, manufacturing, people? Human vs animal pharma, from the companies' own SEC filings.